

Costa Rica — Substation ESG Report v4.2

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks, across the 169-substation Costa Rica fleet. Updated monthly via the SSI Index v4.2 scoring engine (101 variables · 34 sources · 11 modifiers · Re composite bounded [0.920, 1.787]).

SUBSTATION Planta Hidroelectrica Belen r18480861 · Alajuela, alajuela	SSI SCORE (R_{MEDIAN}) 0.257 ● Medium Risk 0.6th percentile	VOLTAGE CLASS 230 kV High-voltage transmission
CONFIDENCE INTERVAL 0.204 – 0.300 P5–P95 · CI width 0.096 · high confidence	MARKOV ETTC 35 yr Expected time to criticality	DER PENETRATION 0.07 T1 transition stress score

Component Fingerprint + Re composite v4.2

Six-component weighted decomposition of the SSI composite score (C / V / I / E / S / T) — the substation's structural risk profile — PLUS the v4.2 **Re composite**, a bounded resilience aggregate of the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just). Layout: *left* — the 6 components rendered as a weighted contribution bar · *right* — the same 6 components rendered as horizontal bars side-by-side with Re_norm as the comparable 7th bar (terracotta) on the [0, 1] scale · *below* — the Re composite tier card (numeric value + bounded trajectory bar + classification + formula explainer). Re composite re-appears as the 7th axis of the ESG Radar in Section A below.

WEIGHTED CONTRIBUTION TO R_{BASE}

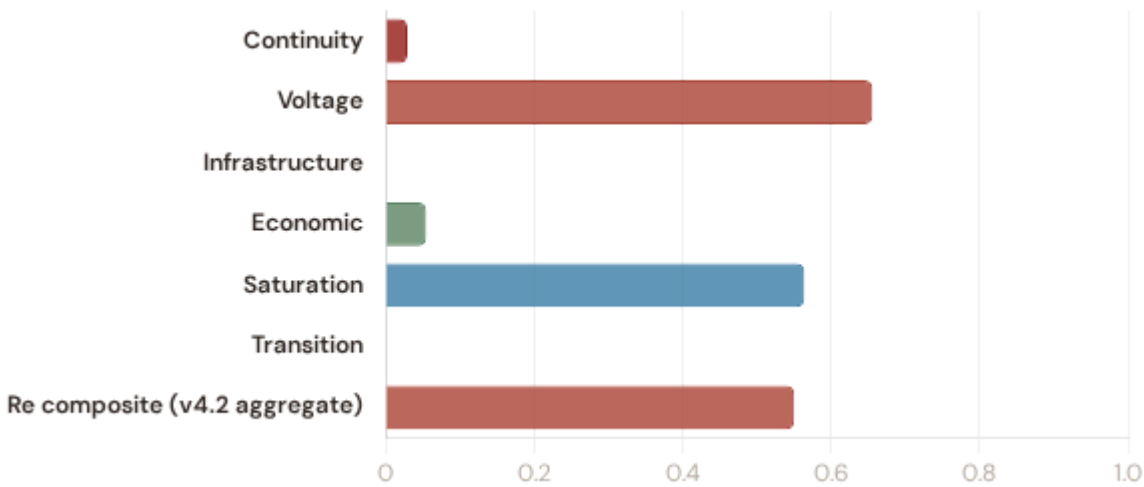


V4.2 RE COMPOSITE (AGGREGATE, NOT IN R_{BASE}) elevated exposure



■ Continuity **0.030** ×0.3 ■ Voltage **0.656** ×0.1 ■ Infrastructure **0.000** ×0.25 ■ Economic **0.055** ×0.1
■ Saturation **0.564** ×0.2 ■ Transition **0.000** ×0.05 ■ **Re composite v4.2 aggregate 0.550** agg

■ **R_{base} total** (weighted sum) **0.193** $\Sigma = 1.00$



0.550
RE_NORM

ELEVATED EXPOSURE

LOW MODERATE ELEVATED (1.00)
(0.00) (0.25) (0.50)



Re_raw 1.1677 (bounds [0.920, 1.787]; normalised to [0, 1])

FORMULA

$$Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$$

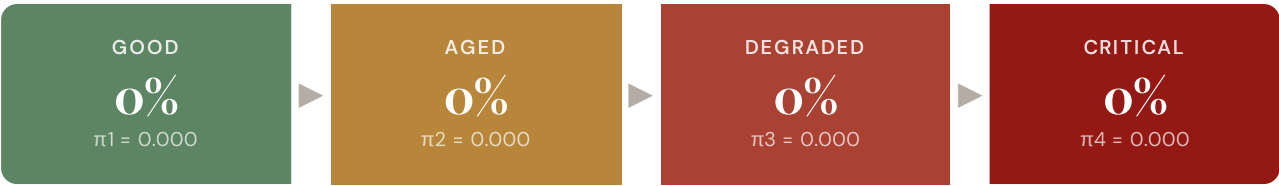
Bounded resilience aggregate of the v4.2 family. $Re < 1.0$ when R8 adaptive capacity dominates other near-neutral factors.

Re composite tier classification: **low** ($Re_norm < 0.25$) · **moderate** ($0.25-0.50$) · **elevated** (≥ 0.50). Bounds [0.920, 1.787] map to Re_norm [0, 1] via $\text{clip}((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)$.

Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

STEADY-STATE DISTRIBUTION



Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

RISK SCORE

28.300

Composite Markov risk

ETTC

35 yr

Expected time to critical

P(CRITICAL) 20YR

17.0%

Probability of reaching critical

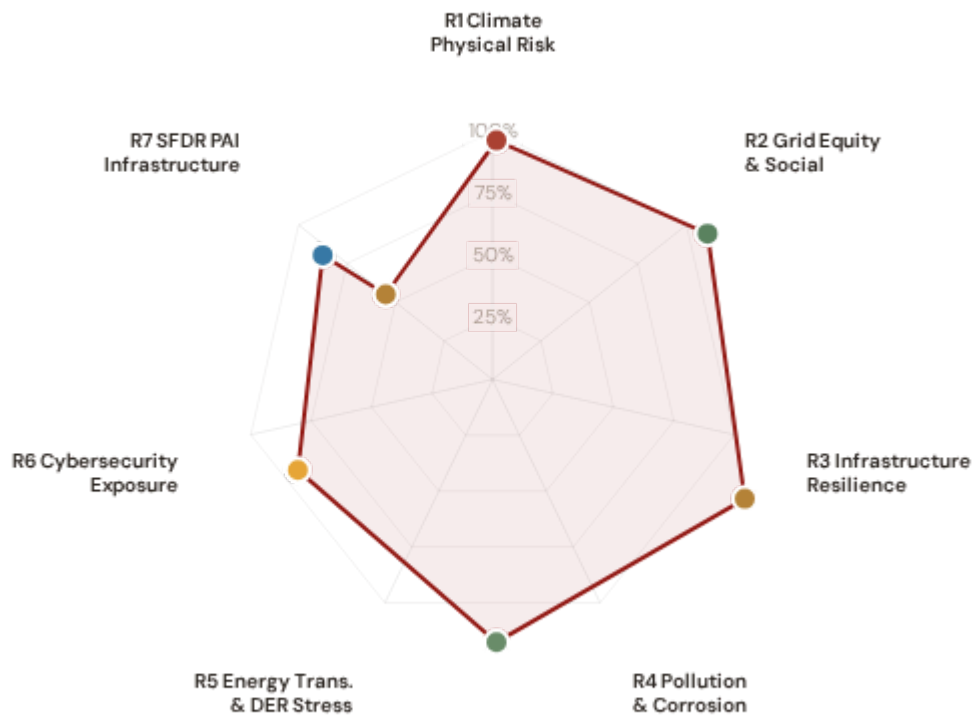
CORROSION

C3

ISO 9223 classification

ESG Radar — Seven-Report Overview ^{v4.2}

A Composite readiness across 7 ESG dimensions per FC v3 §14 (R1 Climate Physical · R2 Grid Equity & Social · R3 Infrastructure Resilience [Re composite ESG home] · R4 Pollution & Corrosion · R5 Energy Transition & DER Stress · R6 Cybersecurity · R7 SFDR PAI Infrastructure, FC v3 §14 subsection 13.7), each linked to UN Sustainable Development Goals. *Note: R7 here is the SFDR ESG-disclosure axis (Re_norm proxy under Convention #7) — distinct from the v4.0.2 formula-chain R7 cyber/digital modifier documented in methodology.html. The two R7s share a number by design (V4_2_IMPLEMENTATION_ARCHITECTURE.md §4.5).*



Re composite v4.2 7th axis · Re_raw 1.1677 · Re_norm 0.550 · ELEVATED EXPOSURE



R1 · Climate Physical Risk Assessment PARTIAL

SDG 13 Climate Action · SDG 9 · SDG 11

SDG 13

SDG 9

SDG 11

R2 · Grid Equity & Social Vulnerability READY

SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1

SDG 7

SDG 10

SDG 1

R3 · EU Taxonomy Alignment READY

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12

SDG 9

SDG 13

SDG 12

R4 · Energy Transition & DER Stress READY

SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7

SDG 13

SDG 9

R5 · Pollution & Corrosion READY

SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15

SDG 11 SDG 3 SDG 15

● **R6 · Cybersecurity Exposure** **PARTIAL**

SDG 9 Industry, Innovation, Infrastructure · SDG 16

SDG 9 SDG 16

● **R7 · SFDR PAI Infrastructure** **GAP** **v4.2**

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 11 · $Re_norm = 0.550$ (elevated exposure)

SDG 9 SDG 13 SDG 11

C **Audit Trail & Methodology Note**
Traceability for CSRD Article 29a limited assurance

ITEM	VALUE
Scoring Engine	SSI Index v4.2 — Gaussian copula Monte Carlo (10,000 iterations) · 11 modifiers (5 baseline + 6 v4.2 resilience family) · Re composite bounded [0.920, 1.787]
Weight Architecture	C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05
Modifiers Applied	R3 C_mult: 1.022, R4 F_topo: 1.000, R6 restoration: 0.900, R6 seismic: 1.000, R6 volcanic: 1.000, R6 hurricane: 0.020, R6 hydro_deficit: 0.100, R7 cyber: 1.025, R10 just: 1.035, R6d wildfire: 1.052, R6c flood: 1.055, R8 adapt: 0.980, R9 compound: 1.037, R6e winter: 1.006
Degradation Model	Markov 5-state, CIGRE TB 761 calibration
Thermal Model	IEEE C57.91 transformer loading curves
Report Date	17 March 2026
Reporting Period	Calendar year 2025
Refresh Cadence	Annual (data ingestion → scoring engine → display)
Substation	Planta Hidroelectrica Belen (r18480861)
Data Assurance Level	All inputs from institutional open-source with citable vintage
Re composite (v4.2)	Re_raw = 1.1677 · Re_norm = 0.550 · ELEVATED exposure · bounds [0.920, 1.787]

D

Re composite — bounded resilience aggregate (v4.2)

Resilience composite anchoring CSRD/ESRS climate-resilience disclosure

The Re composite (v4.2) is a bounded resilience aggregate capturing the new R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just modifier family in a single scalar suitable for CSRD/ESRS climate-resilience disclosure:

$$\text{Re_raw} = (\text{R6d} \times \text{R6e} \times \text{R8} \times \text{R9} \times \text{R10}) + (\text{R6c} - 1.00)$$

Bounds: **[0.920, 1.787]**. Re_raw < 1.000 when R8_adapt < 1.00 dominates other near-neutral factors. Costa Rican fleet observed range across 169 substations: the methodology-bounded extremes [0.920, 1.787] — low-end values arise at interior low-exposure substations with R8_adapt < 1.00 dominating near-neutral other modifiers, high-end values at coastal compound-exposure substations where R6c flood + R9 compound stack multiplicatively. v4.2 explicitly retires the pre-Path-B "winter exposure is a real climate burden" reading per Convention #6 §2.3.1 (Bourgouin freezing-rain p99 tail-event amplifier per §6 design — winter exposure signal correctly lives in v4.0.2 I1 Snow/Ice metric, where it methodologically belongs). See [W1-W10 FC v2](#) for per-substation worked examples + W4 Climate resilience baselines.

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.